

CONTACT

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SKILLS

R (Data Analysis) 8+ yrs
(tidyr,dplyr)

R (Visualisation) 8+ yrs
(ggplot2,plotly)

R (Report Writing) 5+ yrs
(rmarkdown,quarto)

Develop R Packages 8+ yrs
(devtools,usethis, roxygen2,testthat, covr, pkgdown, drat)

R (Dashboard) 5+ yrs
(Shiny,flexdashboard, golem,DT)

Latex and MS Office 8+ yrs

Spatial Analysis 1+ yrs
(ArcGIS and QGIS)

SAS and SQL 3 yrs

Python, C++ and Java 2 yrs

Teaching 2 yrs

Presentation 5 yrs

AMALAN MAHENDRAN

Data Scientist with a PhD in Statistical Theory

EDUCATION

PhD in Statistics

Feb, 2020 - Dec, 2025

School of Mathematical Sciences, Faculty of Science,
Queensland University of Technology, Australia.

Title - New Methods in Experimental Design for Subsampling Big Data.

B.Sc. (Hons) in Statistics,

Mar, 2013 - Dec, 2016

Department of Statistics and Computer Science,
Faculty of Science, University of Peradeniya, Sri Lanka.

CGPA with 3.87/4.00 CGPA.

Subjects: Statistics, Mathematics, Computer Science.

Bayesian Statistical Learning by Dr. Matias Quiroz

Jan, 2022 - Feb, 2022

Summer School in the Mathematical Sciences,
Australian Mathematical Science Institute.

WORK EXPERIENCE

Research Fellow in Transport Engineering and Planning

April, 2025 - ongoing

School of Civil and Environmental Engineering,
Faculty of Engineering,
Queensland University of Technology, Australia.

Conducting crash safety analysis using state-of-the-art modelling tools.
Responsible for data cleaning, preparation, and evaluation of suitable analytical methods based on project requirements.

Teaching Assistant and Tutor

Jul, 2024 - Nov, 2025

School of Mathematical Sciences, Faculty of Science,
Queensland University of Technology, Australia.

Help students to successfully handle the industry-collaborated project.
Mark team work logs, presentation and modelling reports.
MXB 348 - Work Integrated Learning in Statistics.

Temporary Lecturer

Jan, 2017 - Jul, 2018

Department of Statistics and Computer Science,
Faculty of Science, University of Peradeniya, Sri Lanka.

Prepare and conduct lectures and practicals.

Set and mark exam papers, quizzes, tutorials.

CS 100 - Computer Applications (Practical) - Instructor (2017/2018).

ST 103 - Statistics Applications 1 - Instructor and In-charge (2017).

ST 104 - Statistical Applications 2 - Instructor and In-charge (2017).

ST 204 - Sampling Techniques - Tutor (2017).

ST 205 - Statistical Simulation - Tutor (2017).

CS 202 - Data Structures Practical - Tutor (2017/2018).

ST 301 - Regression Analysis - Instructor (2017/2018).

ST 305 - Multivariate Methods I - Tutor (2018).

ST 407 - Linear Models - Tutor (2017).

AS 404 - Data Integrity Management and Data Analysis - Instructor (2018).

AWARDS

**Australian Government
Research Training
Program Stipend and fees
offset (International).**

(Queensland University of
Technology, Australia)

**Presidents Awards for
Scientific Research
2019-2021.**

(National Research Council
Sri Lanka)

**Department of Statistics
and Computer Science
Alumni Prize for
Excellence in Statistics.**

(University of Peradeniya,
Sri Lanka)

**University Prize for
Academic Excellence.**

(University of Peradeniya,
Sri Lanka)

PUBLICATIONS

Mahendran, A., Thompson, H., & McGree, J. M. (2023).A model robust subsampling approach for Generalised Linear Models in big data settings. *Statistical Papers*, 1-21.

Published ([Link](#))

Amalan Mahendran, Pushpakanthie Wijekoon, (2019). fitODBOD: An R Package to Model Binomial Outcome Data using Binomial Mixture and Alternate Binomial Distributions. *Journal of Open Source Software*, 4(39), 1505.

Published ([Link](#))

Ozim, C. O., Mahendran, R., Amalan, M., & Puthussery, S. (2023). Prevalence of human immunodeficiency virus (HIV) among pregnant women in Nigeria: a systematic review and meta-analysis. *BMJ open*, 13(3), e050164.

Published ([Link](#))

Mahendran R, Puthussery S, Amalan M. Prevalence of antenatal depression in South Asia: a systematic review and meta-analysis. *J Epidemiol Community Health* Published.

Published ([Link](#))

Mahendran, A., Thompson, H., & McGree, J. M. (2025). A subsampling approach for large data sets when the Generalised Linear Model is potentially misspecified.

Published ([Link](#))

**NeEDS4BigData: An R Package for New Experimental
Design Based Subsampling Methods for Big Data**

Writing

Mahendran, A., Thompson, H., & McGree, J. M.

CONFERENCES & WORKSHOPS

**Transport Academic Partnership Showcase 2025
Brisbane, Queensland, Australia**

3 December, 2025

Poster: Predicting and Mapping Expected Crashes to Support Road Safety Planning in Southeast Queensland

**Model-Oriented Data Analysis and Optimum Design
Conference 2023
Southampton, United Kingdom**

8 - 15 Jul, 2023

Presentation: A model robust subsampling approach for Generalised Linear Models in big data settings

**Australasian Applied Statistics Conference 2022
Victoria, Australia**

28 Nov - 2 Dec, 2022

Poster: A model robust design approach for optimally subsampling big data

R PACKAGES

NeEDS4BigData:
Implementing
subsampling methods to
find informative samples
from big data.

R package version 1.0.1.

**fitODBOD: Modeling Over
Dispersed Binomial
Outcome Data Using BMD
and ABD.**

R package version 1.5.4.

**fitODBODRshiny: Shiny
Application for R Package
"fitODBOD".**

R package version 1.0.2.

SLPresElection:
**Presidential Election Data
of Sri Lanka from 1982 to
2015.**

R package version 1.0.0.

**OlympicRshiny: Shiny
Application for Olympic
Data.**

R package version 1.0.2.

**SouthParkRshiny: Data
and Shiny Application for
the Show "SouthPark".**

R package version 1.0.0.

EXTRACURRICULAR

- Consulting Rajarata University to prepare the **statistics curriculum for Data Science** (2019).
- **Article Reviewer** of Rajarata International Research Conference (2019).
- Successfully completed a Three-day induction program conducted for Temporary Academic staff, by the **Staff Development Centre**, University of Peradeniya (2017).
- Member of the **Data Science Unit** of Post Graduate Institute of Science, University of Peradeniya (2017-2018).
- Member of the **Media Publicity Committee** for the Annual Research Congress 2017 (Rescon2017), 2017.
- **Organising Committee member** of Industry Day 2017, Faculty of Science, University of Peradeniya (2017).
- Organised activities for **"Statistical Camp 2017"**, Department of Statistics and Computer Science, Faculty of Science, University of Peradeniya (2017).
- Organised a short introduction session explaining statistics and its applications for school students at the "Faculty Open Day-2017- Faculty of Science, University of Peradeniya" (2017).
- Team member for conducting **one-day Statistical Seminars** for Advanced level Students in Mathematics Stream (2016).
- Group leader and an active member and winning group from **"Statistics Camp 2015"** conducted by the Department of Statistics and Computer Science (2015).

REFERENCES

Prof. James McGree,
Professor in Statistics,
School of Mathematical Sciences,
Faculty of Science,
Queensland University of Technology,
Brisbane, QLD, 4000.
E-mail - james.mcgree@qut.edu.au
Phone - +61-7-3138-2313

Prof. Helen Thompson
Professor in Statistics,
School of Mathematical Sciences,
Faculty of Science,
Queensland University of Technology,
Brisbane, QLD, 4000.
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